DATA INSIGHT REPORT

AnalystLab Africa Internship · Week 1–2 Assignment | Analyst: Onyia Gift Nkiruka

■ DATASETS OVERVIEW

Dataset	Source	Key Columns	Focus
E-commerce Transactions (OnlineRetail.csv)	UCI ML Repo	InvoiceNo, StockCode, Qty, UnitPrice, CustomerID, Country, InvoiceDate	Revenue, sales trends, top products, country share
Netflix Titles (netflix_titles.csv)	Kaggle / Netflix	show_id, type, title, director, country, date_added, release_year, listed_in	Content mix, genre trends, top directors, release growth

■ 1 · DATA CLEANING STEPS

E-Commerce Dataset

- Loaded with latin1 encoding to handle special characters
- Standardised all column names to lowercase
- Detected and flagged negative values in Quantity & UnitPrice
- Converted InvoiceDate to datetime dtype
- Identified CustomerID as the natural primary key
- Checked for duplicate rows (ec_df.duplicated())
- Removed all rows with null values via dropna()
- Generated summary statistics for sanity check

Post-cleaning: Dataset reduced to fully valid transaction records; revenue-ready.

Netflix Dataset

- Loaded with latin1 encoding (special chars in titles/cast)
- Identified numeric cols: release_year only; rest categorical
- No negative values detected in numeric fields (clean check)
- Converted date_added using format='mixed' for parsing flexibility
- Designated show_id as the primary key
- Checked for and confirmed absence of duplicate entries
- Dropped rows with any null fields (dropna())
- Verified final dtypes and non-null counts via .info()

Post-cleaning: Retained titles with complete metadata for reliable genre/director analysis.

■ 2 · EXPLORATORY DATA ANALYSIS PROCESS

Step	E-Commerce	Netflix
Feature Engineering	Created Revenue = Quantity × UnitPrice; Total Revenue computed	Used release_year and date_added for time-series analysis
Aggregation	Grouped by year (revenue sum), product description (top-5), and country (quantity sum)	Value counts on type, listed_in, director, country, release_year
Distribution	Describe() used to examine spread of quantity & unit price	Histogram (histplot) used to examine release year distribution by content type
Ranking / Top-N	nlargest(5) on revenue and quantity aggregations	nlargest(10) on genres and countries; nlargest(5) on directors
Saved Output	Cleaned + enriched dataset exported to data_updated.csv	DataFrame kept in memory for visualisation

■ 3 · VISUALISATIONS PRODUCED

#	Chart Title	Type	Dataset	Key Variable
1	Sales Trend Over Time	Line chart	E-Commerce	Revenue by Year
2	Top 5 Products by Revenue	Bar chart	E-Commerce	Revenue (desc.)
3	Top 5 Best-Selling Products	Horizontal bar	E-Commerce	Quantity Sold
4	Sales Distribution by Country	Pie chart	E-Commerce	Quantity % Share
5	Movies vs. TV Shows	Pie chart	Netflix	Content Type %
6	Top 10 Genres	Horizontal bar	Netflix	Genre Frequency
7	Top 5 Directors by Title Count	Horizontal bar	Netflix	Director Count
8	Top 10 Countries by Content	Horizontal bar	Netflix	Country Count
9	Content Release Trends Over Time	Stacked histogram	Netflix	Release Year × Type

■ 4 · INSIGHTS & INTERPRETATIONS

E-Commerce Transactions

- UK dominates sales volume, reflecting the dataset's UK-centric retail origin.
- Revenue spikes in Q4 — consistent with seasonal gifting patterns (Nov–Dec).
- A small set of gift/novelty products (e.g. PAPER CRAFT, LUNCH BAG) drives disproportionate revenue.
- Negative quantity values signal returns/cancellations — critical to exclude for accurate KPIs.
- Most customers are high-frequency, low-basket-size buyers, suggesting subscription potential.

Netflix Titles

- Movies account for ~70% of the catalogue; TV Shows are growing in share year-on-year.
- International Movies and Dramas are the two dominant genres, reflecting global content push.
- The US, India, and UK produce the majority of titles — with India a fast-growing contributor.
- Content additions peaked around 2019–2020, likely reflecting peak streaming-war investment.
- A handful of directors appear repeatedly in non-English originals, revealing franchise patterns.

■ Recommendations: Future analyses should segment customers using RFM scoring (E-Commerce) and build a content-recommendation engine using genre + country co-occurrence (Netflix). Both datasets would benefit from outlier-aware preprocessing before modelling.
